ECEN 5613 Final Project: 8051 MIDI control surface for Reaper (Hardware_Map.md Rev 13.4)
Authors: Nathan Cohen and Travis Nagle. One sheet per board section; nets join across sheets by global label.

Class board: 8051, address latch, decode, reset, status LED

File: class_mcu.kicad_sch

Class board: Power, motor supply, motor driver and direction latch

File: class_power.kicad_sch

Class board: Serial, USB-MIDI, I2C and panel cables

File: class_io.kicad_sch

D-pad board: Button buffer, fader ADC, D-pad, LCD contrast

File: dpad.kicad_sch

Fader button board: Fader buttons F1/F2 and touch sensors

File: fader_strip.kicad_sch

Panel Mounted: Motor faders, LCD and pedal jack (wired to the boards)

File: panel.kicad_sch

Button daughter board: PCF8574 buttons and LEDs on the RJ45 I2C bus (Phase 3)

File: btn_daughter.kicad_sch

Inter-board connections:
J20 (class) <-> J30 (D-pad)   12-pin ribbon, about 9 in: AD0-AD7, LCD_RS, LCD_RW, LCD_E, GND
J21 (class) <-> J31 (D-pad)   10-pin ribbon, about 8 in: +5V, GND x2, BTN_OE_N, TOUCH2, ADC SPI, 1 spare
J22 (class) --> FDR1, FDR2 motors   4 wires (two twisted pairs)
J32 (D-pad) <-> J40 (fader strip)   6 wires: +5V, GND, BTN_F1, BTN_F2, TOUCH1, TOUCH2
J33 (D-pad) --> LCD1   14 wires; J34 (D-pad) --> FDR1/FDR2 pots   4 wires; J35 (D-pad) --> J13 pedal jack   3 wires
J41 (fader strip) --> FDR1/FDR2 touch pins T   2 wires
J11 (class) RJ45 --> J101 (button daughter) patch cable; J102 --> next daughter board
Power: J1 (class) <-- 9 V >= 2 A adapter. USB-C to the KB2040 (A2). DB9 J5 / J7 TTL debug to the PC.

Authors: Nathan Cohen, Travis Nagle (AI-assisted, see report)
Generated from hardware/kicad/tools/design.py by gen_schematic.py
Spec: Hardware_Map.md Rev 13.4
ECEN 5613 Final Project

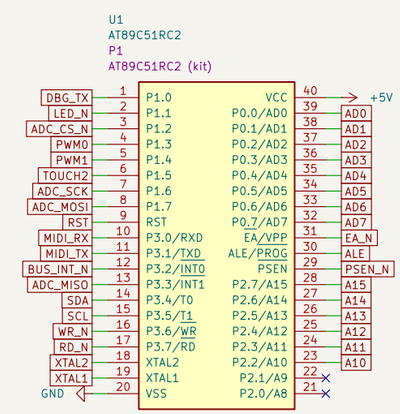
Sheet: /
File: ESD_Final.kicad_sch

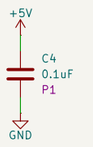
Title: Reaper MIDI control surface: boards and cables

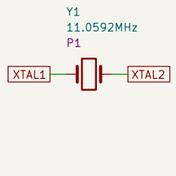
Size: A3	Date: 2026-10-25	**Rev: 13.4**
KiCad E.D.A. 10.0.6		Id: 1/8

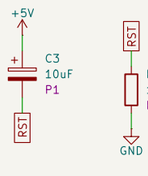
Existing class development board circuit (reference/class_board), with the project changes:

- U4 NVSRAM removed and returned (no external memory); A8, A9 and latch outputs O2–O7 now unused.
- U7 reprogrammed with spld/final.pld; new wire-wrap: A11 -> pin 9, A10 -> pin 11, outputs 12, 14, 15.
- U3 O0/O1 tapped for LCD_RS / LCD_RW. P1.0–P1.7, P3.2–P3.5 are new project signals.
- SW4 (/INT0 button) now shares BUS_INT_N with the daughter-board interrupt.

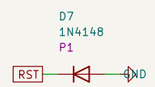




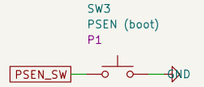


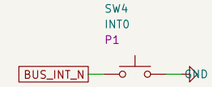


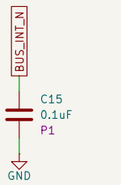


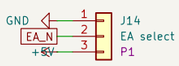


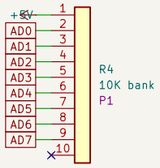


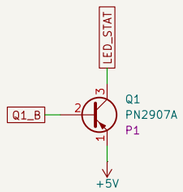




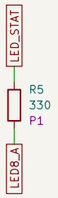


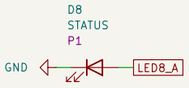


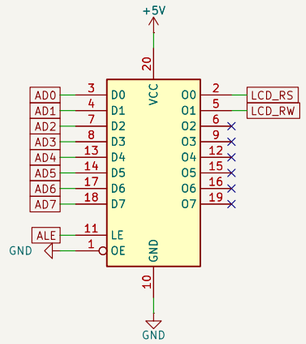


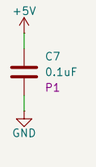


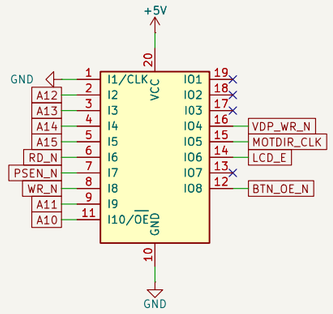


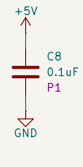


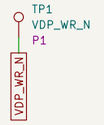














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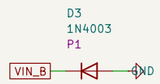
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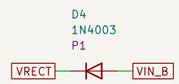
Power: one 9 V >= 2 A adapter (Hardware_Map 3.9). VM is tapped at the bridge output (VRECT).
SW1 sits before the bridge, so it now carries motor current too: VERIFY its rating (>= 2 A) or replace it with a toggle switch rated >= 3 A. A1 GND returns to the bridge '−' output (star point).
Bottom-level additions (A1, U9, U14, Q2, F1, D9, C16–C21, J22) are built on the class board.

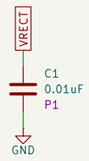


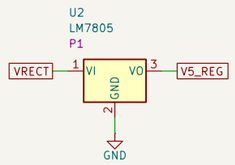


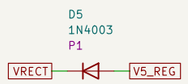


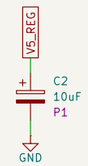


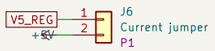


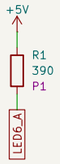


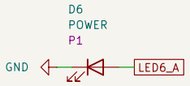


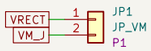


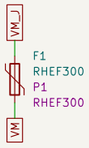




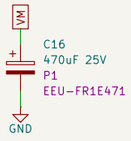


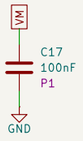


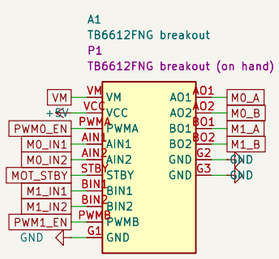




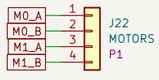


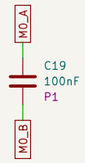


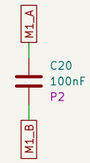


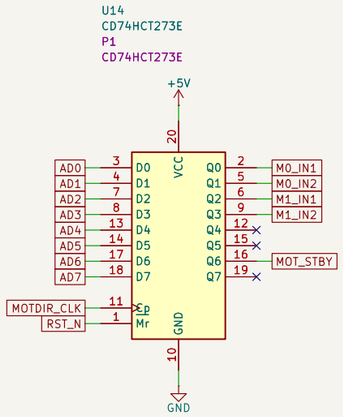


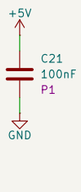


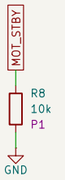


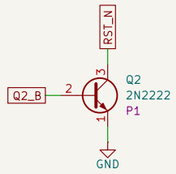


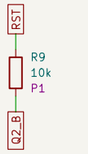


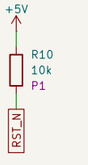


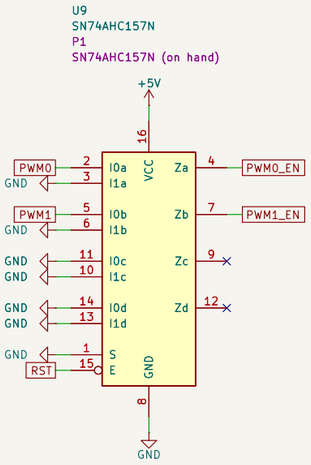


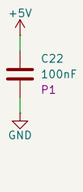














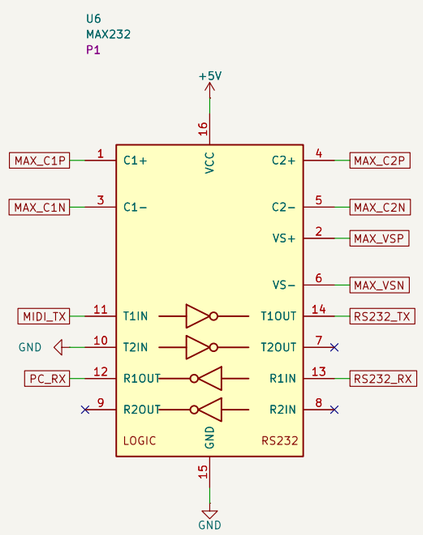
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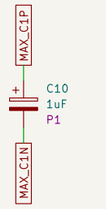
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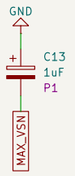
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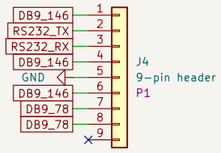
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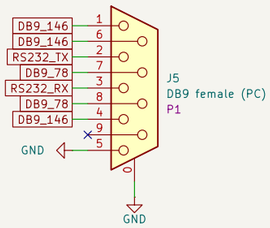
JP2 (JP_RX): 1–2 = USB-MIDI (KB2040), 2–3 = PC / FLIP bootloader through the MAX232 and DB9.
A2 KB2040 is powered from its own USB-C; its RAW/5V is not joined to +5V.
J20 (12-pin) and J21 (10-pin) are the class-board ends of the panel ribbon cables (to J30 / J31).

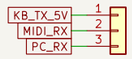




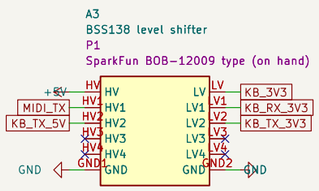


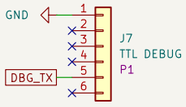


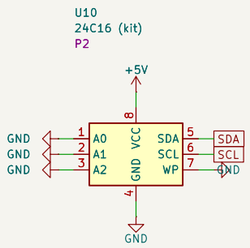




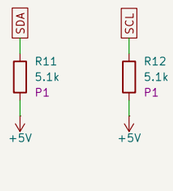


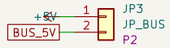


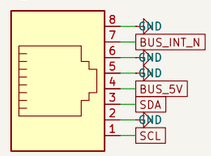


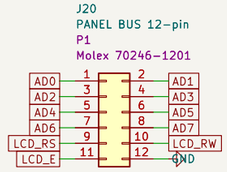


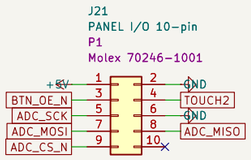














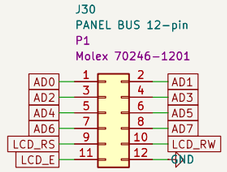
Authors: Nathan Cohen, Travis Nagle (AI-assisted, see report)
Generated from hardware/kicad/tools/design.py by gen_schematic.py
Spec: Hardware_Map.md Rev 13.4
ECEN 5613 Final Project: Reaper MIDI control surface

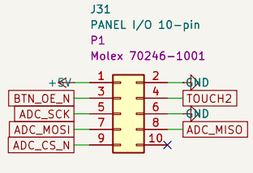
Sheet: /Class board: Serial, USB-MIDI, I2C and panel cables/
File: class_io.kicad_sch

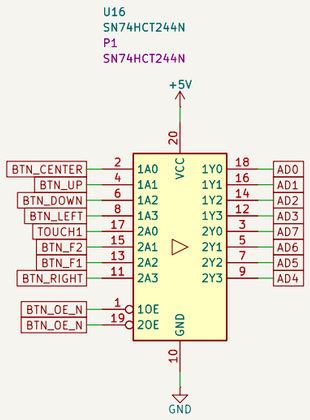
Title: Class board: Serial, USB-MIDI, I2C and panel cables

Size: A3	Date: 2026-10-25	**Rev: 13.4**
KiCad E.D.A. 10.0.6		Id: 4/8

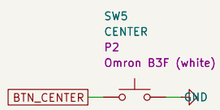
Panel level. J30 / J31: D-pad-board ends of the panel ribbon cables (from class board J20 / J21).
J32 -> fader strip J40; J33 -> LCD1; J34 -> fader pots; J35 -> pedal jack J13.
TOUCH1 (U16 D7) has no pull-up: the AT42QT1010 output is push-pull (10 nF filter on the fader strip).
AGND (GNDA) joins GND only at NT1, next to U11.

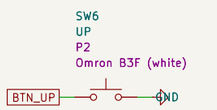


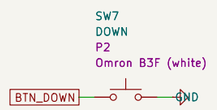


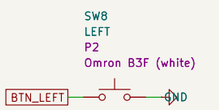


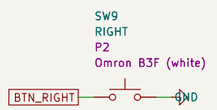


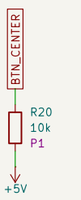


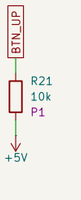


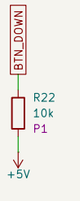


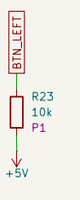


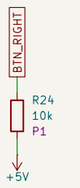


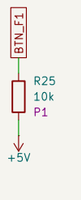


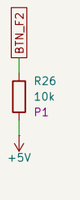


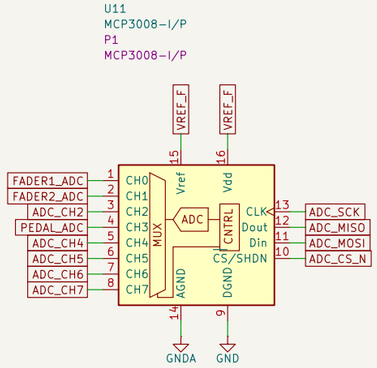


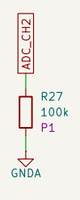


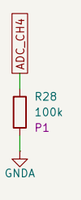


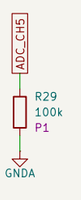


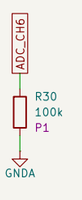


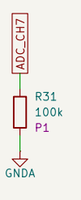


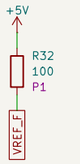


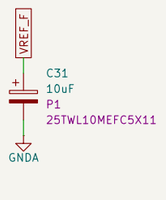


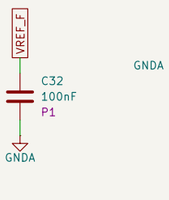




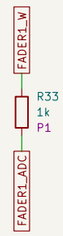


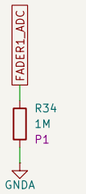


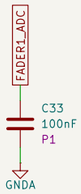


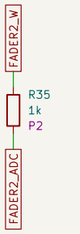


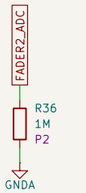


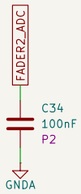


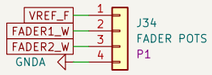


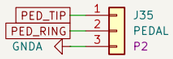


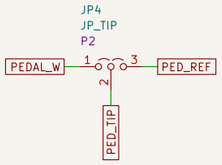


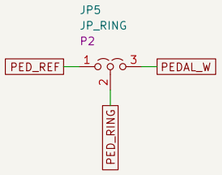


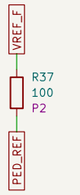


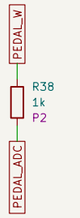


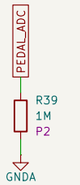


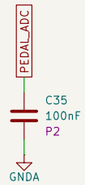


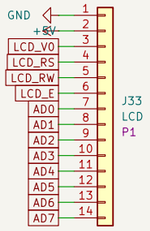


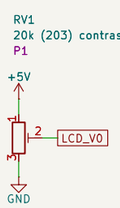


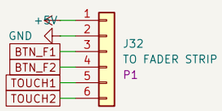
















Authors: Nathan Cohen, Travis Nagle (AI-assisted, see report)
Generated from hardware/kicad/tools/design.py by gen_schematic.py
Spec: Hardware_Map.md Rev 13.4
ECEN 5613 Final Project: Reaper MIDI control surface

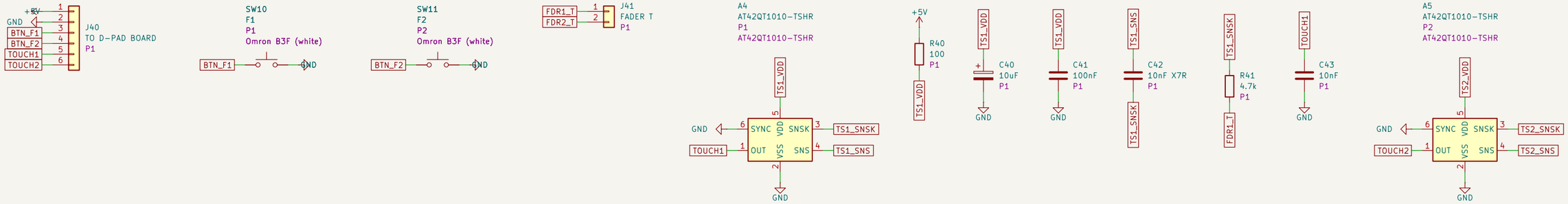
Sheet: /D-pad board: Button buffer, fader ADC, D-pad, LCD contrast/
File: dpad.kicad_sch

Title: D-pad board: Button buffer, fader ADC, D-pad, LCD contrast

Size: A3	Date: 2026-10-25	**Rev: 13.4**
KiCad E.D.A. 10.0.6		Id: 5/8

Panel level. J40 -> D-pad board J32. J41 -> fader touch pins T (FDR1, FDR2), wires about 2 cm.

AT42QT1010: SYNC/MODE to GND = low-power mode (about 60 s hold limit). Cs tune 2-50 nF.



Authors: Nathan Cohen, Travis Nagle (AI-assisted, see report)
Generated from hardware/kicad/tools/design.py by gen_schematic.py
Spec: Hardware_Map.md Rev 13.4
ECEN 5613 Final Project: Reaper MIDI control surface

Sheet: /Fader button board: Fader buttons F1/F2 and touch sensors/
File: fader_strip.kicad_sch

Title: Fader button board: Fader buttons F1/F2 and touch sensors

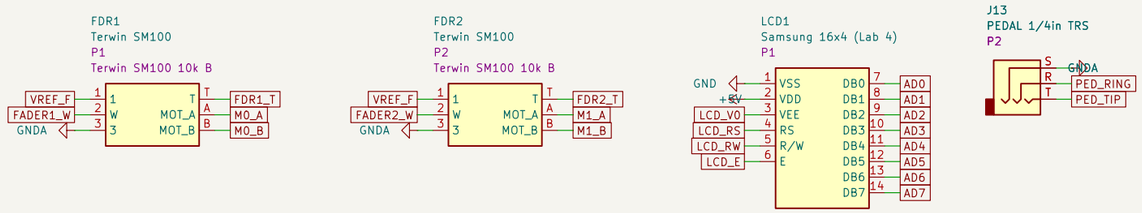
Size: A3	Date: 2026-10-25	**Rev: 13.4**
KiCad E.D.A. 10.0.6		Id: 6/8

Parts mounted on the acrylic panel, wired to the boards (hardware/panel):

FDR1/FDR2 pots -> D-pad J34; T -> fader strip J41; motors A/B -> class board J22.

LCD1 -> D-pad J33 (14 wires). J13 pedal jack -> D-pad J35.

VERIFY the fader wiper pin (drawing suggests pin 2) before wiring.



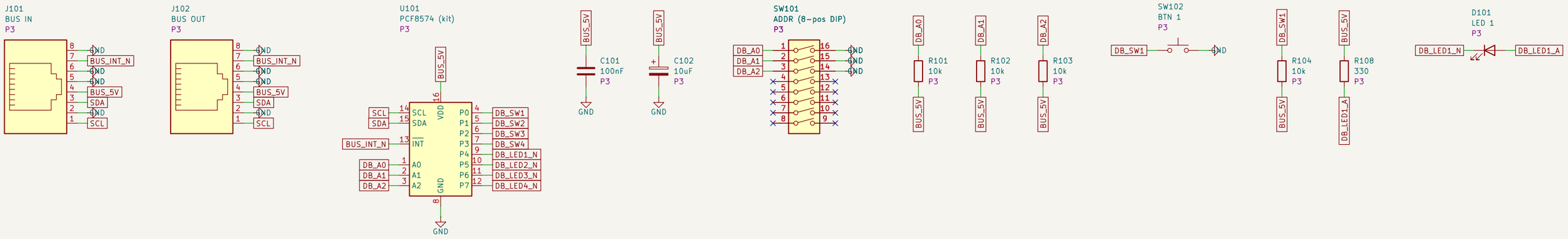
Authors: Nathan Cohen, Travis Nagle (AI-assisted, see report)
Generated from hardware/kicad/tools/design.py by gen_schematic.py
Spec: Hardware_Map.md Rev 13.4
ECEN 5613 Final Project: Reaper MIDI control surface

Sheet: /Panel Mounted: Motor faders, LCD and pedal jack (wired to the boards)/
File: panel.kicad_sch

Title: Panel Mounted: Motor faders, LCD and pedal jack (wired to the boards)

Size: A3	Date: 2026-10-25	**Rev: 13.4**
KiCad E.D.A. 10.0.6		Id: 7/8

Phase 3 daughter board on the RJ45 I2C bus (J101 bus in, J102 bus out, wired in parallel).
SW101 positions 1–3 set A0–A2 (closed = 0): I2C address 0x20 + n. Positions 4–8 unused.
Powered from BUS_5V (class board +5V through JP3 / JP_BUS).



Authors: Nathan Cohen, Travis Nagle (AI-assisted, see report)
Generated from hardware/kicad/tools/design.py by gen_schematic.py
Spec: Hardware_Map.md Rev 13.4
ECEN 5613 Final Project: Reaper MIDI control surface
Sheet: /Button daughter board: PCF8574 buttons and LEDs on the RJ45 I2C bus (Phase 3)/
File: btn_daughter.kicad_sch

Title: Button daughter board: PCF8574 buttons and LEDs on the RJ45 I2C bu

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